

MOTIVATION

Human Action Recognition is essential for Human-Robot Collaboration (HRC) in manufacturing. Robots must understand human behavior to operate safely and efficiently. Recognizing worker actions helps improve task coordination, reduce accidents, and enable autonomous decision-making.

PROBLEM DEFINITION:

Industrial HRC requires reliable recognition of worker actions from video, yet complex backgrounds, motion blur, and subtle movements make single-modality methods unreliable. Our project solves this by extracting human skeletons with YOLOv8-Pose and learning structural motion via ST-GCN, while a parallel 3D CNN captures RGB appearance and temporal dynamics. A late-fusion strategy integrates both cues to produce robust action classification.

PROBLEM CHALLENGE:

Challenge in HRC Video	Why It Causes Failure	Our Solution
Cluttered industrial background	RGB model overfits to irrelevant objects	Add Skeleton Stream focusing only on human joints
Pose occlusion / blur	Keypoints become noisy	RGB stream complements skeleton with visual context
Fine-grained motions	Subtle arm/hand moves look similar in RGB	ST-GCN models body topology + temporal dynamics
Lighting / viewpoint variation	RGB appearance unstable	Skeleton representation is lighting-invariant
Uncertain predictions	Single modality unreliable	Late fusion merges both cues for robust output

METHODOLOGY

Skeleton Branch

- YOLOv8-Pose extracts 17 keypoints per frame; sequences are resampled to 64 frames.
- An ST-GCN models human joint topology (spatial) and motion dynamics (temporal).
- Outputs skeleton-based action probabilities $p_{skeleton}$.

RGB Branch

- Uniformly sample 16 frames, resize to 112×112, and form a 3D video clip.
- A 3D ResNet-18 (r3d_18) pretrained on Kinetics-400 captures appearance and motion.
- Produces RGB-based probabilities p_{rgb} .

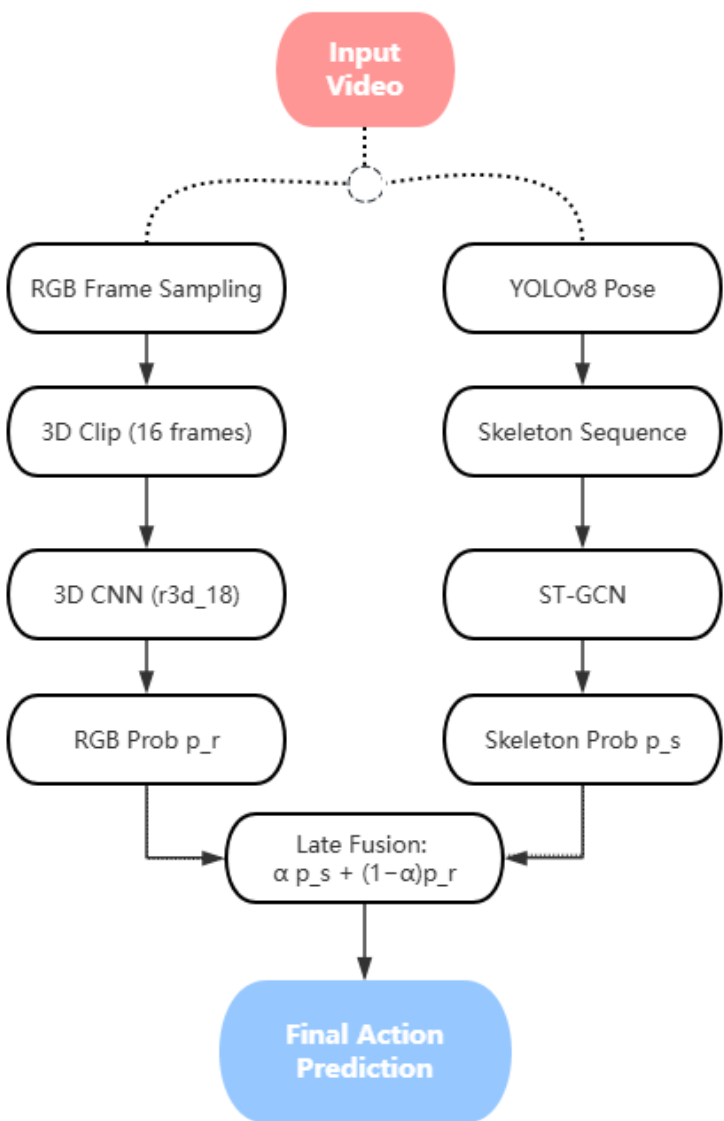
Late Fusion

Final prediction is computed by weighted score fusion:

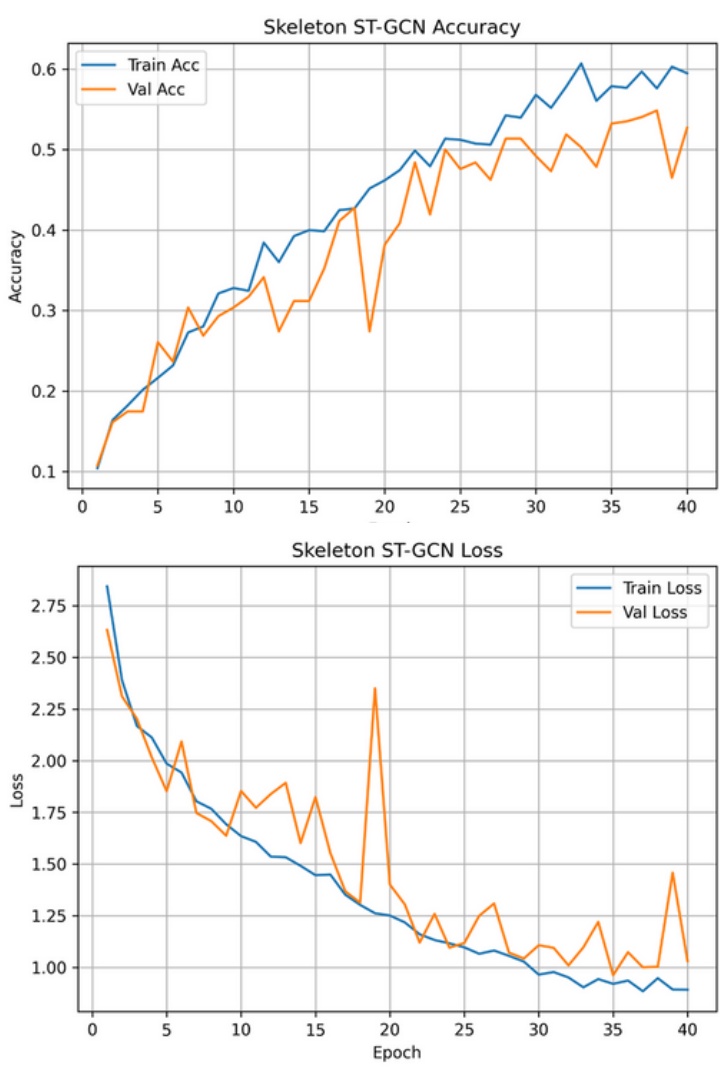
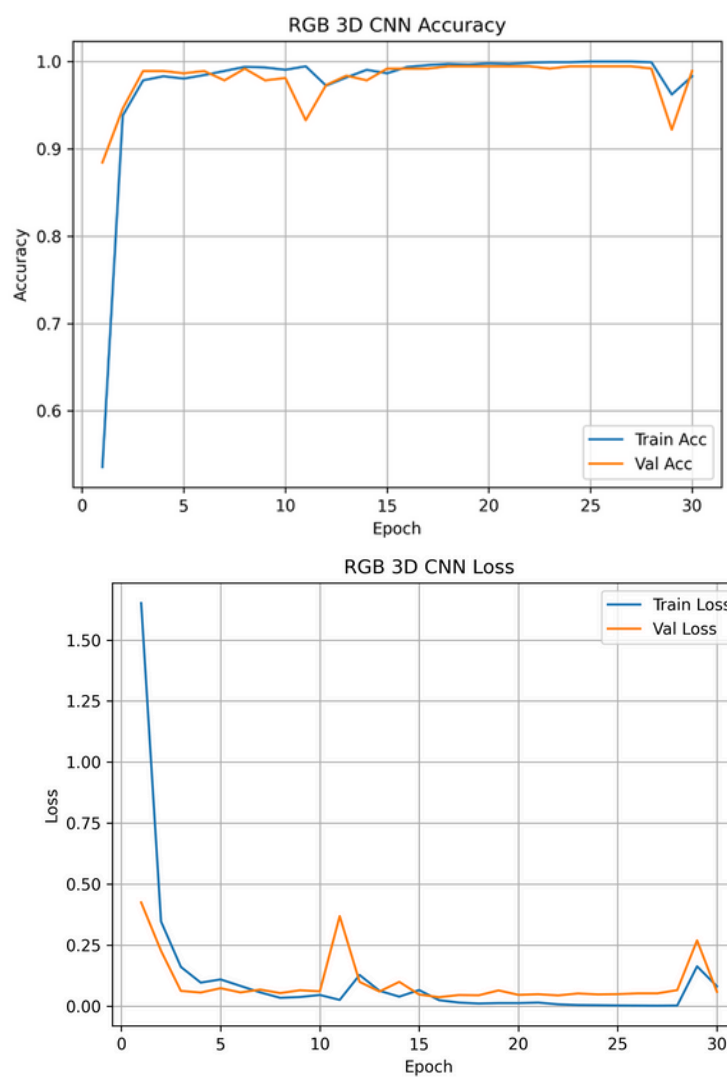
$$p_{final} = \alpha p_{skeleton} + (1-\alpha)p_{rgb}, \quad \alpha=0.6$$

Implementation

- **Training:**
 - ST-GCN: Adam, lr=1e-3, batch 32, 40 epochs.
 - RGB r3d_18: Adam, lr=1e-4, batch 8, 30 epochs.
- **Hardware:** Single NVIDIA GPU used for training and pose extraction.
- **Inference:** Both branches output class probabilities, which are fused to generate the final action label.



EXPERIMENTAL RESULTS



- RGB: Fast convergence, near 100% accuracy, strong visual features.
- Skeleton: Slower improvement, 50–60% accuracy, adds essential motion
- Accuracy of final test set (Combining the two models): 71.15%

Model Curve Observation	Explanation
RGB Loss drops sharply, accuracy approaches 99%	Visual model is very strong and easy to train
Skeleton Loss decreases slowly, final accuracy 50–60%	Actions are harder; skeleton information is sparse but still effective
Two branches complement each other	Fusion can improve overall representation

CONCLUSION

We have completed the design of the HRC hybrid action recognition system and achieved an accuracy of 71.15% in the test set. After publishing the labeled test set and modifying the weight coefficients of the skeleton model, the credibility of RGB was improved, and the final accuracy was significantly improved. The reason for this may be that the skeleton model is too simple.

FUTUREWORK

Based on previous learning explorations and corresponding achievements, there are three tasks that need to be further expanded in the future:

1. Use more complex models to analyze the skeleton and explore whether the accuracy of the training set in this section will be significantly improved.
2. Adjust the ratio of the training set and validation set to adjust the model training, in order to verify and analyze the performance of the model under different training volumes.
3. Explore the performance of skeleton and RGB under more complex background and environmental conditions.

REFERENCE

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